

VINAY DILIP JAYASELAN

Energy Consultant | Digital Twin | VDC Engineer | BIM Specialist | Architect

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PROFESSIONAL SUMMARY

Professional with a multidisciplinary background in energy-efficient building design, digital twins, and advanced BIM workflows. Experienced in developing IoT-enabled platforms for real-time building performance monitoring, integrating sensor networks, MQTT communication, SQL databases, and Python-based analytics. Skilled in Revit and Rhino.Inside, Grasshopper, and data-driven modeling, with a strong foundation in building physics, HVAC, and energy modeling. Recognized for delivering precise models, coordinated outputs, and automation-focused solutions that improve building operations.

KEY SKILLS

Revit | AutoCAD | Rhino | Rhino.Inside | Grasshopper | Navisworks | Power BI | Python | SQL | JavaScript | WUFI | Linux | AWS | Adobe Suite | MS Office

PROFESSIONAL EXPERIENCE

VDC – BIM Engineer

Doxel.ai → ModelyzeLabs.ai, California (Hybrid)

January 2022 – April 2025

(Team transitioned internally from Doxel.ai to ModelyzeLabs.ai during restructuring)

- Led VDC strategy implementation and spearheaded BIM adoption, contributing to BEPs and improving project efficiency and model accuracy.
- Managed 3D model development, coordination, and version control using Revit, Navisworks, Colibri, and Dynamo for modeling, visualization, and clash detection.
- Implemented technology-driven workflows to enhance constructability, reduce rework, and support interdisciplinary coordination with project managers and design teams.
- Performed BIM-based quantity take-offs, cost estimation, schedule integration, and applied construction methodology knowledge to optimize VDC processes across project types.
- Delivered BIM/VDC outputs for complex project types, including hospitals, pharmaceutical facilities, data centers, and battery-manufacturing facilities.

BIM Architect

GMSC Consultant, Bangalore, India

October 2020 – July 2021

- Orchestrated multidisciplinary project coordination across design, modeling, and interference checks, ensuring alignment between architectural, structural, and MEP disciplines.
- Managed design and construction workflows for major hospital and educational institution projects, delivering client presentations during and after project completion.
- Produced accurate 2D drawings from BIM models, handling sheet management, annotations, dimensions, notes, visibility settings, and compliance with BIM standards and family creation.
- Provided on-site engineering support and coordinated drawings, layouts, and technical requirements with PHE, Electrical, and Fire teams while assisting clients throughout project delivery.

Architect

Rathi Associate, Bangalore, India

July 2019 – September 2020

- Developed and produced Revit content, including detailed design elements, schedules, and building-services installation documentation.

- Communicated technical issues clearly to contractors, consultants, and clients, maintaining strong attention to quality and detail throughout project delivery.
- Applied working knowledge of AutoCAD, Revit, and 3D modeling to support coordinated design and documentation workflows across disciplines.
- Demonstrated strong proficiency in model and drawing production, ensuring accuracy and consistency across Revit and AutoCAD deliverables.
- Worked on commercial buildings, residential apartment projects, and township planning assignments

Architect - Intern

B+A Architects, Bangalore, India

July 2018 – July 2019

- Supported design and construction phases for national-level projects, preparing drawing layouts and coordinating with clients.
- Delivered research, technical detailing, and documentation for F&B, residential, retail, and commercial projects

Architect - Intern

Citati Architects, Bangalore, India

June 2014 – March 2018

(Internship completed alongside Bachelor of Architecture studies)

- Produced construction drawings, architectural details, revised plans, and 3D models aligned with project requirements.
- Assisted small-scale residential and commercial projects with architectural documentation and visual illustrations.
- Worked on residential buildings, schools, and public buildings.

EDUCATION

Ms. Energy-Efficiency in Building Design

Lund University, Lund, Sweden

Graduated: 2026

Thesis: Developing a Cost-Effective Digital Twin Framework for Energy-Efficient Operation in Existing Buildings.

Designed an IoT-enabled digital twin platform integrating sensor networks, MQTT, SQL databases, Python-based data processing, and dashboard visualization for real-time building performance monitoring and energy optimization.

Coursework: Building Energy Design • HVAC • Lighting • Solar Energy • LCA/LCC • Moisture & Heat Analysis • Material Physics

Bachelor of Architecture

VTU, India

Graduated: 2019

Coursework: Architectural Design • Urban Design & Landscape • Construction • Structures & Installations

Diploma In Civil Engineering

RJS Polytechnic, India

Graduated: 2013

Coursework: Construction • Structures & Installations • Estimation & Quantity

CERTIFICATIONS

Revit Architecture | AutoCAD | Python | Rhino.Inside | Grasshopper | EQUA IDA ICE | Power BI

HOBBIES

Photography | Cycling | Sketching | Reading | Travel